

A SHARP DICHOTOMY FOR BOUNDED-GAP SUMSETS

JOHAN LAND

ABSTRACT. Let $k \geq 3$ and $1 \leq d_1 < d_2$. We prove that Erdős Problem 1112 has an affirmative answer exactly when $d_2 \geq k + 1$; in that range the explicit lacunarity ratio $d_2 + 2$ works. Below the threshold we prove more: however rapidly prescribed successive ratios grow, one sequence meets the k -fold sumset of every admissible bounded-gap sequence. The proof combines a reciprocal Beatty construction, Kneser's density theorem, and an elementary proof of the sharp finite subset-sum lemma required at the boundary.

For a set $A \subset \mathbb{N}$, write

$$kA := \{a_1 + \cdots + a_k : a_1, \dots, a_k \in A\},$$

with repetitions allowed. A sequence $A = \{a_1 < a_2 < \cdots\}$ is (d_1, d_2) -admissible if $d_1 \leq a_{n+1} - a_n \leq d_2$ for every n . We address the formulation recorded as Erdős Problem 1112 [1], inspired by a remark of Erdős and Graham [2]. The earlier Lean-verified proof [5] gave the bound $192d_2$. This version improves that bound and replaces finite certificates with an elementary proof.

I am deeply grateful to Stijn Cambie for his substantial contributions to simplifying the proof and for providing the manuscript on which this shortened presentation is based.

Theorem 1 (Sharp dichotomy). *Let $k \geq 3$ and $1 \leq d_1 < d_2$ be integers.*

- (1) *If $d_2 \geq k + 1$, every sequence of positive integers $B = \{b_1 < b_2 < \cdots\}$ with $b_{n+1} \geq (d_2 + 2)b_n$ admits a (d_1, d_2) -admissible A such that $kA \cap B = \emptyset$.*
- (2) *If $d_2 \leq k$, no fixed ratio works. More strongly, for every sequence $R_1, R_2, \dots$ of positive integer lower bounds there is one increasing sequence B of positive integers with $b_{n+1} \geq R_n b_n$ which meets kA for every (d_1, d_2) -admissible A .*

Thus the answer is yes exactly when $d_2 \geq k + 1$.

Above the threshold, a reciprocal Beatty sequence avoids a prescribed lacunary sequence. Below it, every admissible walk has a congruence-class tail in its k -fold sumset; a diagonal construction then meets every such sumset. The sharp finite input is Theorem 2.2, proved in the appendix.

1. THE EXISTENCE CONSTRUCTION

Put $D = d_2$ and assume $D \geq k + 1$.

Lemma 1.1 (Lacunary interpolation). *Let $I = [u, v] \subset (0, 1)$ have length $\ell > 0$, let J be a nondegenerate closed bounded real interval, and let positive integers $q_1 < q_2 < \cdots$ satisfy $q_{n+1} \geq \rho q_n$. If*

$$\ell(\rho - 1) \geq 1, \tag{1}$$

then, after deleting finitely many q_n , some $\theta \in J$ satisfies $\{q_n \theta\} \in I$ for every remaining n .

Proof. For a positive integer q , the set

$$E_q = \{\theta : \{q\theta\} \in I\} = \bigcup_{m \in \mathbb{Z}} [(m + u)/q, (m + v)/q]$$

has components of length ℓ/q , with left endpoints spaced by $1/q$. For some N , the interval $[q_N \inf J - u, q_N \sup J - v]$ has length $q_N |J| - \ell \geq 1$, so it contains an integer. Thus a component K_N of E_{q_N} lies in J . If K_n is a component of E_{q_n} , the allowable indices of components of $E_{q_{n+1}}$ contained in K_n form a real interval of length

$$q_{n+1}|K_n| - \ell = \ell(q_{n+1}/q_n - 1) \geq 1.$$

It contains an integer, giving $K_{n+1} \subseteq K_n$. The nested nonempty compact intervals have a common point θ , as required. $\square$

Proposition 1.2 (Existence). *If $D \geq k + 1$, the ratio $D + 2$ works.*

Proof. Given B with $b_{n+1} \geq (D + 2)b_n$, put

$$\varepsilon = \frac{1}{2D(D+1)}, \quad I = [\varepsilon, \varepsilon + 1/(D+1)], \quad J = [1/D, 1/D + \varepsilon/(D-1)].$$

Lemma 1.1 applies with equality in (1). It yields $\theta \in J$ and N_0 such that

$$\{b_n \theta\} \in I \quad (n \geq N_0). \quad (2)$$

Put $\alpha = 1/\theta$. Then

$$D - 1 < \alpha \leq D, \quad (3)$$

and, since $k \leq D - 1$,

$$1 - k\theta \geq 1 - (D - 1) \max J = 1/D - \varepsilon = \varepsilon + 1/(D + 1) = \max I. \quad (4)$$

Choose an integer $m \geq 1$ with $k \lfloor \alpha m \rfloor > \max_{n < N_0} b_n$, reading the empty maximum as zero, and set

$$A = \{\lfloor \alpha(m + i) \rfloor : i = 0, 1, 2, \dots\}.$$

Its gaps are $D - 1$ or D , hence admissible, and its least k -fold sum exceeds every discarded term. If a remaining b were a sum of k such floors, then for an integer S (the sum of their indices) $b \leq \alpha S < b + k$, whence

$$b\theta \leq S < (b + k)\theta.$$

But $0 < \{b\theta\} \leq 1 - k\theta$ by (2)–(4), so this half-open interval contains no integer. This is a contradiction. $\square$

Remark 1.3. The argument works for every real ratio $\rho > 1 + D/(D - k)$: take J sufficiently close to $1/D$ and a closed interval $I \subset (0, 1 - k \sup J)$ of length $1/(\rho - 1)$.

2. TOOLS FOR TAIL COVERING

Call A *k-tail-covering* if, for integers $q \geq 1$, $0 \leq r < q$, and X_0 ,

$$\{x \geq X_0 : x \equiv r \pmod{q}\} \subseteq kA. \quad (5)$$

2.1. The density shortcut and the finite input. For $X \subseteq \mathbb{N}_0$ write $\underline{d}(X) = \liminf_{N \rightarrow \infty} |X \cap [1, N]|/N$.

Lemma 2.1 (Kneser shortcut). *Let $P = \{p_0 < p_1 < \dots\}$ with $p_0 = 0$. If $p_n \leq cn + C$ eventually, for constants $0 < c < k$ and C , then kP contains a congruence-class tail.*

Proof. For large N , take $n = \lfloor (N - C)/c \rfloor$. Then $p_n \leq N$, so $\underline{d}(P) \geq 1/c > 1/k$. The lower-density consequence of Kneser's theorem [4, 3] needed here is that either $\underline{d}(kP) \geq k\underline{d}(P)$ or kP contains a congruence-class tail. The first alternative is impossible because density is at most one, proving the claim. $\square$

For a finite multiset S , write $\text{SS}(S) = \{\sum_{x \in T} x : T \text{ is a submultiset of } S\}$.

Theorem 2.2 (Finite interval lemma (SHARP)). *If G is a finite set of positive integers with $|G| \geq 3$, $\gcd(G) = 1$, and $M = \max G$, some multiset of at most $M - 1$ elements of G has M consecutive subset sums.*

The elementary proof in Appendix A uses no search or certificate list.

2.2. A slot lemma. For a gap word $g_1, g_2, \dots$, put $p_0 = 0$ and $p_n = g_1 + \dots + g_n$. A tail of the walk is a translate of $P = \{p_n : n \geq 0\}$.

Lemma 2.3 (Slots). *Suppose all gaps belong to a finite G , every member recurs infinitely often, $|G| \geq 3$, $\gcd(G) = 1$, and $M = \max G \leq k$. Then kP contains every sufficiently large integer.*

Proof. By SHARP choose $S = \{\delta_1, \dots, \delta_m\}$, $m \leq M - 1 \leq k - 1$, with $[c, c + M - 1] \subseteq \text{SS}(S)$. Choose distinct occurrences $g_{n_j} = \delta_j$. In summand slot j choose p_{n_j-1} or p_{n_j} . These slots realize a fixed base plus every offset in $[c, c + M - 1]$. Park $k - 1 - m$ summands at $p_0 = 0$. The last summand is a dial p_n . For a large target x , the residual $x - \text{base} - p_n$ decreases to $-\infty$ in jumps at most M . Its first value at most $c + M - 1$ is at least c , so the slots supply the residual. This represents x in kP . $\square$

3. EVERY SUBCRITICAL WALK IS TAIL-COVERING

Proposition 3.1. *For $k \geq 3$, every increasing sequence of positive integers with gaps at most k is k -tail-covering.*

Proof. Discard a finite prefix so that the gaps lie in the nonempty tail alphabet G of values recurring infinitely often. Put $p_0 = 0$ and $p_n = g_1 + \cdots + g_n$. Divide all gaps and prefix sums by $g = \gcd(G)$. A class tail modulo g rescales to one modulo gg ; translating back only changes the residue. Relabel so that $\gcd(G) = 1$. If $M = \max G < k$, then $p_n \leq Mn$, and Lemma 2.1 applies. Thus assume $M = k$. If $|G| \geq 3$, use Lemma 2.3. The remaining alphabet is $G = \{\delta, k\}$ with $1 \leq \delta < k$ and $\gcd(\delta, k) = 1$. Set $e = k - \delta$, encode gap k by $h_n = 1$ and gap δ by $h_n = 0$, and write $q_n = h_1 + \cdots + h_n$. Then

$$p_n = \delta n + e q_n. \quad (6)$$

For $s \geq 0$ set

$$W_s = \left\{ \sum_{j=1}^k q_{i_j} : i_j \geq 0, \sum_{j=1}^k i_j = s \right\}.$$

This is an integer interval $[w^-(s), w^+(s)] \cap \mathbb{Z}$: compositions are connected by unit transfers, each changing the value by $-1, 0$, or 1 . A path between extremizers misses no intermediate value. Adding or deleting one unit shows that each endpoint rises by 0 or 1 as s increases by one. Write $\text{width}(s) = w^+(s) - w^-(s)$.

Sweep. If $\text{width}(s) \geq k - 1$ eventually, then kP contains every large integer. Indeed, a target x is represented at budget s exactly when $w(s) = (x - \delta s)/e \in W_s$. The integral values of $w(s)$ occur in one class of s modulo e . Along this class w drops by δ per step, whereas either endpoint rises by at most e . Since $w^+(s) \leq s$, for large x the target starts above the band at the first admissible budget beyond the width threshold; eventually it becomes negative. If it skips the band, consecutive admissible budgets satisfy

$$w^+(s) < w(s) = w(s + e) + \delta \leq w^-(s + e) - 1 + \delta \leq w^-(s) + e - 1 + \delta.$$

Thus $\text{width}(s) \leq k - 2$, a contradiction.

A binary word is *balanced* if equal-length factors differ in their numbers of ones by at most one. Define

$$V(\sigma) = \max_{i+j=\sigma} (q_i + q_j) - \min_{i+j=\sigma} (q_i + q_j).$$

If $V(\sigma) \leq 1$ for every σ , the word is balanced: comparing factors with endpoint indices i, i' and j', j , where $i + j = i' + j'$, bounds their count difference by $V(i + j)$. Let $z_n = n - q_n$ count zeros. Then

$$p_n = kn - e z_n. \quad (7)$$

If $\liminf z_n/n > 0$, choose $0 < \gamma \leq 1$ with $z_n \geq \gamma n$ eventually; then $p_n \leq (k - e\gamma)n$, and the density shortcut applies. Assume $\liminf z_n/n = 0$. There are infinitely many σ with $V(\sigma) \geq 2$. Otherwise a sufficiently late tail would be balanced by the same endpoint comparison. Since zero recurs, it has a length- L factor containing two zeros. Every length- L factor would then contain a zero, giving $z_n \geq n/L - O(1)$, a contradiction.

If $k = 2u + 1$, fix σ_0 with $V(\sigma_0) \geq 2$. For all large s , split the slots into u pairs of subtotal σ_0 and a singleton carrying the rest. Independent variation gives width at least $uV(\sigma_0) \geq k - 1$, so the sweep applies.

For $k = 2u$, $u \geq 2$, choose $\sigma_0 < \sigma_1$ with $V(\sigma_i) \geq 2$, and put $\Delta = \sigma_1 - \sigma_0$. Suppose at a sufficiently large s the width is at most $k - 2$, and put $\tau = s - (u - 1)\sigma_0$. Using $u - 1$ pairs of subtotal σ_0 and one of subtotal τ gives width at least $k - 2 + V(\tau)$, so $V(\tau) = 0$. Using $u - 2$ pairs of subtotal σ_0 , one of subtotal σ_1 , and one of subtotal $\tau - \Delta$ likewise gives $V(\tau - \Delta) = 0$. Now $V(n) = 0$ implies $q_i + q_{n-i}$ is constant. Successive differences give $h_{i+1} = h_{n-i}$, so the prefix of length n is a palindrome. The prefixes of lengths τ and $\tau - \Delta$ are therefore palindromes; the shorter is also a suffix of the longer, which consequently has period Δ . Infinitely many width failures would give such periodic prefixes of unbounded length, forcing the whole word to have period Δ . A period contains a zero, contradicting $\liminf z_n/n = 0$. Thus width is eventually at least $k - 1$, and the sweep finishes. $\square$

4. THE UNIVERSAL LACUNARY CERTIFICATE

Lemma 4.1. *If every admissible walk is k -tail-covering, then any prescribed positive integer ratios R_n admit one increasing B with $b_{n+1} \geq R_n b_n$ meeting kA for every admissible A .*

Proof. List pairs (q, r) , $q \geq 1$, $0 \leq r < q$, with each recurring infinitely often, by concatenating the finite lists with $q \leq N$ for $N = 1, 2, \dots$. Write the list as (q_n, r_n) . Choose $b_1 > 0$ in its prescribed class and let b_{n+1} be the least integer in the next class exceeding $\max\{R_n b_n, b_n, n\}$. The sequence grows as required. For any A , its class in (5) occurs infinitely often; the corresponding b_n are unbounded, so one exceeds X_0 and belongs to kA . $\square$

Proof of Theorem 1. Part (1) is Proposition 1.2. For $d_2 \leq k$, Proposition 3.1 and Lemma 4.1 give part (2). $\square$

APPENDIX A. PROOF OF THE FINITE INTERVAL LEMMA

Write u^m for the multiset of m copies of the positive integer u .

A.1. Reduction to a dense triple.

Lemma A.1 (Coprime rectangle). *If $1 \leq p < q$ and $\gcd(p, q) = 1$, then*

$$\text{SS}(p^{q-1}, q^{p-1}) \supseteq [(p-1)(q-1), pq-1].$$

The interval has $p+q-1$ terms, at multiset cost $p+q-2$.

Proof. For n in the interval choose $0 \leq i < q$ with $ip \equiv n \pmod{q}$. Then $j = (n - ip)/q$ satisfies $-1 < j < p$, so $0 \leq j \leq p-1$. $\square$

Lemma A.2 (Interlacing). *Let $d, q \geq 1$, $\gcd(d, q) = 1$, and $q \leq N$. If $[A, A+N-1] \subseteq \text{SS}(C)$, then scaling C by d and adjoining $d-1$ copies of q gives N consecutive subset sums.*

Proof. For $0 \leq n < N$ choose $0 \leq j < d$ with , and put $t = ((d-1)q + n - jq)/d$. Since $0 \leq (d-1)q + n - jq < dN$, we have $0 \leq t < N$. Thus

$$d(A+t) + jq = dA + (d-1)q + n$$

supplies the required interval. $\square$

Lemma A.3 (Two-generator targets). *For coprime $1 \leq p < h$ and $N \geq 2h$, at most $N - \lfloor N/h \rfloor$ terms from $\{p, h\}$ have N consecutive subset sums.*

Proof. Start with the rectangle's interval of length $p+h-1$. Write $N = sh + r$, $s \geq 2$, $0 \leq r < h$, and add $t = s-1 + \mathbf{1}_{\{r \geq p\}}$ copies of h . The translates overlap or abut and cover at least N integers, at cost $h+p+s-3 + \mathbf{1}_{\{r \geq p\}}$. At $s=2$ its slack below $N-s$ is $h+r-p-1 - \mathbf{1}_{\{r \geq p\}} \geq 0$; increasing s increases slack by $h-2$. $\square$

Proposition A.4 (Inductive funnel). *Assuming SHARP below maximum M , it suffices at M to treat triples*

$$G = \{a < b < M\}, \quad \gcd(a, b) = \gcd(a, M) = \gcd(b, M) = 1, \quad a + b \geq M + 2. \quad (8)$$

Proof. Set $H = G \setminus \{M\}$, $d = \gcd(H)$, and $e = \max(H/d)$. If $|H| \geq 3$, induction gives e consecutive subset sums at cost $e-1$ from H/d . Adjoin $\lceil M/e \rceil - 1$ copies of e , obtaining at least M consecutive sums. Scale by d and interlace with $d-1$ copies of M ; $\gcd(d, M) = 1$. The cost is $C = e + d + \lceil M/e \rceil - 3$. Writing $M = de + s$ with $s \geq 1$ gives

$$M - 1 - C = (d-1)(e-2) + s - \lceil s/e \rceil \geq 0,$$

since $e \geq 3$. Otherwise $H = \{a, b\}$, $a < b$. If $d = \gcd(a, b) > 1$, apply Lemma A.3 to $\{a/d, b/d\}$ with target M , scale, and interlace with $d-1$ copies of M . Here $M > b$, $d \geq 2$, and $\lfloor M/(b/d) \rfloor \geq d$, so the cost is at most $M-1$. Now assume $\gcd(a, b) = 1$. If $b \leq M/2$, the same target lemma costs at most $M-2$. If $b > M/2$ and $a+b \leq M$, one more copy of b extends the rectangle to length $a+2b-1 \geq M$, at cost at most $M-1$. If $a+b = M+1$, the rectangle itself suffices. Thus $a+b \geq M+2$ remains. Finally, if $d = \gcd(a, M) > 1$, apply the target lemma to $\{a/d, M/d\}$ with target M , at cost at most $M-d$, scale, and interlace with $d-1$ copies of b ; $\gcd(d, b) = 1$. The case $\gcd(b, M) > 1$ is symmetric. $\square$

For a triple in (8), put

$$e = b - a, \quad h = M - b, \quad \mu = M - a = e + h. \quad (9)$$

Then

$$1 \leq h \leq a - 2, \quad (10)$$

so $a \geq 3$ and $\gcd(a, e) = \gcd(a, \mu) = 1$.

A.2. Paired movers.

Lemma A.5 (Paired movers). *If $3 \leq a < b < M$, $\gcd(a, b) = 1$, and $b + M = ra$, then for*

$$x = r - 1, \quad y = \lceil (M - r)/2 \rceil, \quad z = \lfloor (M - r)/2 \rfloor,$$

$\text{SS}(a^x, b^y, M^z)$ contains M consecutive integers and $x + y + z = M - 1$.

Proof. Here $r \geq 3$, and

$$M \geq a + r - 1, \quad M \leq a(r - 1) - 1 : \quad (11)$$

indeed $2M \geq ra + 1 \geq 2a + 2r - 2$ and $M = ra - b$. Put $p = \lfloor a/2 \rfloor$, $q = \lfloor (a - 1)/2 \rfloor$, and $L = \max(pb, qM)$. Then $p + q = a - 1$, $p \leq y$, and $q \leq z$. For $n \in [L, L + M - 1]$ choose $w \in [0, a - 1]$ with $wb \equiv n \pmod{a}$. If $w \leq p$, use w unpaired copies of b and divide $(n - wb)/a = i + r\kappa$, $0 \leq i < r$. Otherwise put $w' = a - w \leq q$, use w' copies of M , and divide $(n - w'M)/a = i + r\kappa$. The numerators are nonnegative, so $\kappa \geq 0$. Each of κ further pairs (b, M) contributes ra , and $i \leq x$. The copies are available provided

$$L + M - 1 + pM \leq a(ry + x), \quad (12)$$

$$L + M - 1 + qb \leq a(rz + x). \quad (13)$$

For example, in the first branch $a(i + r\kappa) = n - wb$ and $a(i + r(\kappa + w)) = n + wM$; these inequalities bound $\kappa \leq z$ and $\kappa + w \leq y$. The second branch is symmetric. If $L = qM$, the inequalities reduce to $aM - 1 \leq a(ry + x)$ and $qra + M - 1 \leq a(rz + x)$. The latter uses $q \leq z$ and $M - 1 \leq ax$; the former uses

$$ry + x \geq r(M - r)/2 + r - 1 \geq M,$$

since $M \geq a + r - 1 \geq r + 2$. If $L = pb$, the first inequality follows from $p \leq y$ and $M - 1 \leq ax$. The second is $(a - 1)b + M - 1 \leq a(rz + x)$. Odd a gives $p = q$ and $qM > pb$, so take $a = 2v$, $v \geq 2$. Put $\tau = M - vr = vr - b \geq 1$. The slack is

$$D = 2vrz - 2(v - 1)b - 2v + 1 = 2vr(z - v + 1) + 2(v - 1)\tau - 2v + 1.$$

Here $z = \lfloor (r(v - 1) + \tau)/2 \rfloor \geq v - 1$. If $z \geq v$, then $D \geq 2vr - 1 > 0$. If $z = v - 1$, then $\tau \neq 1$ (that would give $z \geq v$), so $\tau \geq 2$ and $D \geq 2v - 3 > 0$. Finally $x + y + z = M - 1$. $\square$

A.3. The symmetric residue path.

Lemma A.6 (Centered frame). *If C has a subset sum in one common interval $[L, U]$ for each residue modulo a , put $x = \lceil (M - 1 + U - L)/a \rceil$. Then $\text{SS}(C \cup a^x)$ contains $[U, L + ax]$, with at least M integers.*

Proof. Subtract the chosen representative of $n \pmod{a}$ from $U \leq n \leq L + ax$. The remainder is a nonnegative multiple of a at most ax . $\square$

Lemma A.7 (Symmetric η argument). *If $a < b < M$ satisfy (10), are pairwise coprime, and $a \nmid b + M$, at most $M - 1$ terms from $\{a, b, M\}$ have M consecutive subset sums.*

Proof. Let p, q be units modulo a , let $\eta \in [0, a - 1]$ represent qp^{-1} , and put

$$t = \min(\eta, a - \eta), \quad Z = \lfloor (a - 1)/t \rfloor, \quad r = a - 1 - tZ, \quad K = t - 1 + Z. \quad (14)$$

For $(p, q) = (b, M)$ we have $\eta \equiv \mu e^{-1} \pmod{a}$. The values $0, -1, 1$ would give respectively $a \mid M$, $a \mid b + M$, and $a \mid h$, so $2 \leq t \leq \lfloor a/2 \rfloor$. If $\eta = t$, write $0 \leq n < a$ as $n = tk + j$. The representatives $jp + kq$ lie in $[0, S]$, where

$$S = \max\{(t - 1)p + (Z - 1)q, rp + Zq\}. \quad (15)$$

If $\eta = a - t$, write the consecutive residues $t - a, \dots, t - 1$ as $n = j - tk$; representatives lie in $[0, S]$ with

$$S = (t - 1)p + Zq. \quad (16)$$

All are subset sums of p^{t-1}, q^Z . The centered frame with $L = 0$ costs

$$\mathcal{B} = K + \lceil (M - 1 + S)/a \rceil. \quad (17)$$

We can also reverse (p, q) , since M is a unit modulo a . Always

$$K \leq \lceil a/2 \rceil. \quad (18)$$

Indeed, for $c = \lceil a/2 \rceil$ the concave function $t(c - t + 2)$ exceeds $a - 1$ at both endpoints of $[2, \lfloor a/2 \rfloor]$, so $\lfloor (a - 1)/t \rfloor \leq c - t + 1$. Since $S \leq KM$, the budget holds whenever

$$(a - K - 1)M \geq a(K + 1) - 1. \quad (19)$$

For $a = 3$ all unit residues are already excluded.

Odd $a = 2n + 1$. If $K \leq n$, the slack in (19) at $K = n$ is $n(\mu - 2) \geq 0$. Furthermore $t - 1 + \lfloor 2n/t \rfloor = n + 1$ forces $t \in \{2, n\}$, since $(t - 2)(t - n) < 0$ for $2 < t < n$. Only $\eta \in \{2, -2, n, n + 1\}$ remains. If $\eta = 2$, then $e - h = \lambda a$ for $\lambda \geq 0$ by (10). Here $Z = n$, $r = 0$, $S = nM$. With $\epsilon = 0$ for $h = 1$ and $\epsilon = 1$ for $h \geq 2$, evaluation gives

$$(M - 1) - \mathcal{B} = n\lambda + h - 2 - \epsilon. \quad (20)$$

This is nonnegative if $\lambda \geq 1$, or if $\lambda = 0$ and $e = h = g \geq 3$. For $g = 1$, take $n + 1$ copies of a , one of $a + 1$, and n of $a + 2$. At selection level $n + 1$, offsets fill $[0, a]$; at the next level offsets $[1, a]$ remain. The intervals abut, of total length $2a + 1 \geq a + 2$. For $g = 2$, take $n + 3$ copies of a , one of $a + 2$, and n of $a + 4$. With $c = n + 1$, each level $\ell \in \{c, c + 1, c + 2\}$ contains $\ell a + 2[0, a]$. On $[(c + 1)a, (c + 3)a]$, the middle level supplies one parity and the other levels supply the other. The costs are respectively $a + 1$ and $a + 3$, as required.

If $\eta = -2$, reverse b, M ; the reversed ratio is n . For $n \geq 3$, the normalized residues $1, \dots, a$ have representatives

$$jM \ (1 \leq j \leq n - 1), \quad b + jM \ (0 \leq j \leq n - 1), \quad 2b, \quad M + 2b.$$

These use M^{n-1}, b^2 and lie in $[b, b + (n - 1)M]$. The centered frame costs $n + 1 + \lceil (nM - 1)/a \rceil \leq M - 1$, since the slack is

$$a(M - n - 2) - (nM - 1) = (n + 1)\mu - 2n \geq 2.$$

For $n = 2$, representatives $M, b, M + b, 2b, M + 2b$ lie in $[b, M + 2b]$. The cost condition is $2e + 3h - 9 \geq 0$. Since $h \equiv 2e \pmod{5}$, pairwise coprimality and $1 \leq h \leq 3$ exclude $e = 1, 2$, so it holds.

For $\eta = n$ or $n + 1$ with $n \geq 3$, use the forward path in the first case and reverse in the second, making its ratio 2. In either case $K = n + 1$ and $S \leq (n - 1)b + M$. The budget is at most $a + 1 + \lceil ((n + 1)e + 2h - 1)/a \rceil$, and its required slack is

$$ne + (a - 2)h - 2a + 1 \geq 0. \quad (21)$$

For $h \geq 2$ this is at least $n - 3$. For $h = 1$, the first orientation gives $a \mid 3e + 2$, so $e \geq 3$; the second gives $a \mid e + 2$, so $e \geq a - 2$. Both imply the inequality.

Even $a = 2n$. If $K \leq n - 1$, (19) holds, with slack $n\mu + 1$ at $K = n - 1$. For $a \geq 12$ and $3 \leq t \leq n - 2$, $t - 1 + \lfloor (2n - 1)/t \rfloor \leq n - 1$. Thus maximal $K = n$ requires $t \in \{2, n - 1, n\}$. As η is a unit, only even n and $\eta = n \pm 1$ remain. For $\eta = n - 1$, (15) gives $Z = 2$, $r = 1$ and $S = (n - 2)b + M$. The slack is

$$ne + 2(n - 1)h - 2n + 1 \geq n - 1. \quad (22)$$

For $\eta = n + 1$, $h \equiv ne \pmod{2n}$; e is odd, so (10) forces $h = n$. Now $S = (n - 2)b + 2M$ and the slack is

$$2n^2 - 7n + (n - 1)e + 1 \geq 2n(n - 3) > 0. \quad (23)$$

For $a = 4, 6$ there are no unit residues beyond ± 1 . For $a = 8$, $\eta = 3$ gives $S = b + 2M$ and slack $8(M - 5) - (3M + b - 1) = 4e + 5h - 7 > 0$. For $a = 8$, $\eta = 5$ forces $h = 4$; $S = 2b + 2M$ gives slack $8(M - 5) - (3M + 2b - 1) = 3e + 5h - 15 > 0$. For $a = 10$, reverse if necessary to make the ratio 3. The height is at most $2b + 2M$, and slack is $10(M - 6) - (3M + 2b - 1) = 5e + 7h - 9 > 0$. All cases succeed. $\square$

Proof of Theorem 2.2. Use strong induction on M , with the statement vacuous for $M < 3$. Proposition A.4 leaves a triple satisfying (8). If $a \mid b + M$, use Lemma A.5; otherwise use Lemma A.7. $\square$

APPENDIX B. FORMAL VERIFICATION AND CORRESPONDENCE

Both directions, including the density dependency and elementary SHARP proof, are checked in Lean. Selected declarations are listed below; paths are relative to `lean/Erdos1112Proof/`.

Paper result	Lean declaration
Existence/Reciprocal.lean	
Lacunary interpolation	<code>reciprocal_interpolation</code>
Existence with ratio $d_2 + 2$	<code>existence_bound_reciprocal</code>
Short/Intervals.lean	
Coprime rectangle	<code>coprime_rectangle</code>
Interlacing	<code>interlacing</code>
Two-generator targets	<code>two_generator_target</code>
Centered frame	<code>centered_frame</code>
Short/Slots.lean	
Slots and final-summand sweep	<code>slots_from_run</code>
Short/Normalization.lean	
Tail alphabet and gcd normalization	<code>normalized_walk</code>
Short/Binary.lean	
Zero-liminf binary width argument	<code>binary_eventual_width</code>
Binary covering from zero liminf	<code>binary_tail_covering</code>
Short/Funnel.lean	
Inductive funnel	<code>sharpAt_of_funnel</code>
Short/Sharp.lean	
Complete finite interval lemma	<code>sharp_all</code>
Short/KneserWeak.lean	
Pairwise density/AP-tail alternative	<code>weak_kneser</code>
Short/Main.lean	
Kneser shortcut	<code>density_shortcut</code>
Subcritical walk covering	<code>all_tailCovering</code>
Universal lacunary construction	<code>strong_nonexistence</code>

The canonical theorems in `Final.lean` use these proofs throughout. The Kneser consequence is proved using e-transforms, residue compression and finite Mann estimates; it is not assumed. The old proofs and certificate tables have been removed. The definitions in `lean/Erdos1112.lean` allow repeated summands, phrase gap bounds additively, and prove equivalence of natural and integer ratio formulations. The repository supplies build and axiom-audit commands.

REFERENCES

- [1] T. F. Bloom, *Erdős Problem #1112*, <https://www.erdosproblems.com/1112>.
- [2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L’Enseignement Mathématique **28**, Université de Genève, Geneva, 1980.
- [3] P. Erdős, M. B. Nathanson, and A. Sárközy, *Sumsets containing infinite arithmetic progressions*, J. Number Theory **28** (1988), 159–166. https://www.renyi.hu/~p_erdos/1988-33.pdf
- [4] M. Kneser, *Abschätzung der asymptotischen Dichte von Summenmengen*, Math. Z. **58** (1953), 459–484. <https://doi.org/10.1007/BF01174162>
- [5] J. Land, *A dichotomy for k -fold sumsets of bounded-gap sequences (Erdős Problem 1112)*, preprint, 2026. <https://doi.org/10.5281/zenodo.21568276>; source and Lean development: https://github.com/beetree/math_erdos_1112.